Modern Lectures on QFT

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Abstract

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1 Scale-Invariant Physics

These notes summarize the Wilsonian renormalization-group viewpoint and its application to the Ising universality class. The prose has been lightly reconstructed; the equations have been normalized and checked for consistency.

1.1 Wilsonian RG in QFT

Consider a Euclidean scalar theory with cutoff Λ:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 + \frac{1}{2}m^2\varphi^2 + \frac{\lambda}{4!}\varphi^4, \tag{1}$$

$$Z = \int [\mathcal{D}\varphi]_\Lambda \exp\left(-\int d^d x \mathcal{L}[\varphi]\right), \quad [\mathcal{D}\varphi]_\Lambda = \prod_{|k|<\Lambda} d\varphi(k), \tag{2}$$

with Fourier convention

$$\varphi(x) = \int \frac{d^d k}{(2\pi)^d} e^{ik\cdot x}\varphi(k). \tag{3}$$

One may worry about changing into the fourier space may cause a change of measure, indeed, yet long as we focus on physical observables, the change of measure does not affect the final results.

> **Remark**
>
> Here in Wilsonian's picture, all QFT are defined up to a cut off in the momentum space or equivalently a lattice spacing in the position space.

1.1.1 Integrating out UV Modes

Now we want to consider how a theory changes as we lower the energy cut off. Now lower the cutoff from Λ to $\rho\Lambda$, where $0 < \rho < 1$, and split

$$\varphi(k) = \varphi_{\text{IR}}(k) + \hat{\varphi}(k), \tag{4}$$

where φ_{IR} has support for $|k| < \rho\Lambda$ and $\hat{\varphi}$ has support for $\rho\Lambda \leq |k| < \Lambda$. Then we assume that m, λ are small and we integrate out the high-momentum modes $\hat{\varphi}$:

$$Z = \int [\mathcal{D}\varphi]_{\rho\Lambda} \int [\mathcal{D}\hat{\varphi}] \exp\left(-\int d^d x \mathcal{L}[\varphi + \hat{\varphi}]\right) = \int [\mathcal{D}\varphi]_{\rho\Lambda} \exp\left(-\int d^d x \mathcal{L}_{\text{eff}}[\varphi]\right). \tag{5}$$

The effective Lagrangian may in principal contains every local operator allowed by the symmetries:

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}(1 + \Delta Z)(\partial\varphi)^2 + \frac{1}{2}(m^2 + \Delta m^2)\varphi^2 + \frac{1}{4!}(\lambda + \Delta\lambda)\varphi^4 + \Delta C(\partial\varphi)^4 + \Delta D\varphi^6 + \ldots \tag{6}$$

All induced coefficients depend on the blocking parameter ρ. Now we see that the original theory if we integrate out the high momentum modes first we generates an equivalent theory with a lower cutoff and new lagrangian. Notice that the new lagrangian is not a completely new theory but rather a effective theory of the original theory if we only consider the low energy physics.

1.1.2 Flow on Coupling Space

Now we want to interpret the new effective theory as a theory with a change of the coupling from the original theory (while preserving the cut off and the Kinetic term). To do this we need to restore the original cutoff by defining

$$x' = \rho x, \quad k' = \frac{k}{\rho}, \quad \partial_\mu = \rho\partial'_\mu. \tag{7}$$

We also want the Kinetic term to be canonically normalized thus we can fix a normalization of the field and only have the coupling to flow.

$$\varphi' = \sqrt{\rho^{2-d}(1+\Delta Z)}\varphi. \tag{8}$$

The action therefore becomes

$$\int d^d x' \left[\frac{1}{2}(\partial'\varphi')^2 + \frac{1}{2}m'^2\varphi'^2 + \frac{\lambda'}{4!}\varphi'^4 + C'(\partial'\varphi')^4 + ...\right], \tag{9}$$

where

$$\begin{aligned} m'^2 &= (m^2 + \Delta m^2)(1+\Delta Z)^{-1}\rho^{-2}, \\ \lambda' &= (\lambda + \Delta\lambda)(1+\Delta Z)^{-2}\rho^{d-4}, \\ C' &= (C + \Delta C)(1+\Delta Z)^{-2}\rho^{d}. \end{aligned} \tag{10}$$

> **Remark**
>
> Notice that until now we have not made any approximation. But obviously the coefficients for the new effective lagrangian are quite impossible to compute.

However, now we focus on leading order in m^2, λ, C and we have:

$$m'^2 = \rho^{-2}m^2, \quad \lambda' = \rho^{d-4}\lambda, \quad C' = \rho^d C. \tag{11}$$

We may see that different couplings behave differently under the RG flow. The mass term m^2 grows as we lower the cutoff, the λ term grows for $d < 4$ and shrinks for $d > 4$, and the C term shrinks for all $d > 0$. Thus we call:

- Relevant operators: m^2 and λ for $d < 4$.
- Irrelevant operators: C and λ for $d > 4$.
- Marginal operators: λ for $d = 4$. In fact this need detailed analysis, in dim reg we in fact work in $d = 4 - \varepsilon$ and we find that λ is marginally relevant for $\varepsilon > 0$.

1.1.3 RG Flow and Fixed Points

The flow of the couplings can be described by a function called the beta function:

$$\mu\frac{dg}{d\mu} = \beta(g). \tag{12}$$

with μ the energy scale and g the coupling. We may find out that there are some fixed point of the RG flow, which have some interesting behaviors.

YL: [Diagram in hand written note!]

1.2 RG in StatMech

For RG in StatMech, it is much more calculable compared to QFT (where in fact we seldome directly calculate the RG flow of Wilsonian coupling.) Now we take Ising model as a simplest model for an illustration. For lattice with spins $s_i \in \{+1, -1\}$ on each site:

$$H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i, \quad P[\{s\}] = \frac{1}{Z}e^{-\beta H[\{s\}]}, \quad Z = \sum_{[\{s\}]} e^{-\beta H[\{s\}]}, \tag{13}$$

where $\beta = \frac{1}{T}$. At $h = 0$, the low-temperature phase has nonzero spontaneous magnetization, while the high-temperature phase is disordered. Near the critical temperature,

$$M \sim (T_c - T)^\beta \tag{14}$$

on the ordered side. Here the exponent β should not be confused with inverse temperature. (This is experimentally true and we will see how to understand this.)

1.2.1 Block Spin Transformation

In RG of QFT we integrate out the high momentum modes and see how the theory changes. Now in SM, it is much convenient to stay in position space and we can make the lattice spacing larger and see how the theory changes. To do this we integrate out the blocks of spins to make it a coarse-grained lattice. This is realized by **Block Spin Transformation**:

$$T\big(s';s_1,...,s_q\big)=\begin{cases}1 \text{ if } s'\sum_{i=1}^{q}s_i>0\\ 0 \text{ otherwise}\end{cases}. \tag{15}$$

This means that if most of the spins in a block are up, then the coarse-grained spin is up, and vice versa or we throw away the contribution.

YL: [Diagram for illustration!]

The coarse-grained Hamiltonian is defined by

$$e^{-\beta H'[\{s'\}]}=\sum_{[\{s\}]}\prod_{\text{blocks}}T\big(s';s_1,...,s_q\big)e^{-\beta H[\{s\}]}. \tag{16}$$

and the new partition function is

$$\begin{aligned}Z'&=\sum_{[\{s'\}]}e^{-\beta H'[\{s'\}]}=\sum_{[\{s'\}]}\sum_{[\{s\}]}\prod_{\text{blocks}}T\big(s';s_1,...,s_q\big)e^{-\beta H[\{s\}]}\\&=\sum_{[\{s\}]}e^{-\beta H[\{s\}]}\sum_{[\{s'\}]}\prod_{\text{blocks}}T\big(s';s_1,...,s_q\big).\end{aligned} \tag{17}$$

Since $\sum_{s'}T=1$, one has $Z'=Z$. We can see that the partition function is never changed, we cut integrate out the blocks first and the integrate out the coarse-grained spins later, which is consist with integrating out the highier momentum modes in QFTs.

One may expect a second step of rescaling to restore the original lattice spacing and the Kinetic term to have the field normalized. However, the Ising CFT is independent of the lattice spacing and the field is already normalized in the definition of the BST to have σ fixed to be $+1$ or -1. Thus we do not need to do any rescaling.

From the new Hamiltonian $H'[\{s'\}]$, reading off the new couplings gives an RG map

$$(J',h',...)=\text{RG}[(J,h,...)]. \tag{18}$$

1.2.2 One-dimensional decimation

For 1D Ising model, we can do the RG transformation exactly. For $d=1$ and $h=0$, choose every two spins to live in a block and integrate out the block with a special BST which only sees the even number blocks and simply ignore the first one. The BST is:

$$T(s';s_1,s_2)=\delta_{s',s_2},\qquad H=-J\sum_i s_is_{i+1}-c\sum_i 1. \tag{19}$$

> **Remark**
>
> Here in the original theory we add a term of $-c$ to the Hamiltonian, which is to nromalize out the vaccum energy.

Summing over inside the blocks gives the coarse-grained Hamiltonian as

$$
\begin{aligned}
e^{-\beta H'[\{s_{\text{even}}\}]} &= \sum_{s_{\text{odd}}} e^{\beta\left[J\sum_{\text{even}}(s_i s_{i+1}+s_{i+1}s_{i+2})+2\sum_{\text{even}} c\right]} \\
&= \prod_{\text{even}} \left[e^{\beta(J(s_i+s_{i+2})+2c)} + e^{\beta(-J(s_i+s_{i+2})+2c)}\right].
\end{aligned}
\tag{20}
$$

Notice that here s_i, s_{i+2} are both fixed number. In the final form, the first term is given by $s_{i+1} = 1$ and the second term is given by $s_{i+1} = -1$. Matching this to $e^{J' s_i s_{i+2} + c'}$, and redefining $K = \beta J$, $K' = \beta J'$, and $C = \beta c$, we have:

$$
K' = \frac{1}{2}\ln(\cosh(2K)), \qquad C' = 2C + \frac{1}{2}\ln(4\cosh(2K)). \tag{21}
$$

With $x = \tanh K$, $0 \le x < 1$,

$$
x' = \mathrm{RG}(x) = x^2. \tag{22}
$$

Thus the unstable zero-temperature fixed point $x = 1$ flows toward the stable infinite-temperature fixed point $x = 0$.

1.3 Linearized flow near a fixed point

Let K_a be a set of couplings and $K'_a = \mathrm{RG}(K_a)$ be the RG transformation. We define a fixed point K^*_a as a point that is invariant under the RG transformation:

$$
K^*_a = \mathrm{RG}(K^*_a). \tag{23}
$$

Let K^*_a be a fixed point. Linearizing the RG transformation gives

$$
K'_a - K^*_a = \left[\frac{\partial K'_a}{\partial K_b}\right]_{K=K^*} (K_b - K^*_b) \equiv T_{ab}(K_b - K^*_b). \tag{24}
$$

Let v^i_a be left eigenvectors (i here is the label of eigenvectors),

$$
v^i_a T_{ab} = \lambda_i v^i_b, \tag{25}
$$

and define scaling variables

$$
u_i = v^i_a (K_a - K^*_a). \tag{26}
$$

We now can see that the RG transformation is diagonalized in the u_i basis: Under one RG transformation, the new scaling variable is

$$
\begin{aligned}
u'_i &= v^i_a (K'_a - K^*_a) \\
&= v^i_a T_{ab}(K_b - K^*_b) + \mathcal{O}\left((K-K^*)^2\right) \\
&= \lambda_i v^i_b (K_b - K^*_b) + \mathcal{O}\left((K-K^*)^2\right) \\
&= \lambda_i u_i + \mathcal{O}(u^2).
\end{aligned}
\tag{27}
$$

Therefore, to linear order around the fixed point, the RG transformation is diagonal in the scaling-variable basis:

$$
u'_i = \lambda_i u_i. \tag{28}
$$

After one blocking step, we commonly use b to denote the linear size of the block and use it to characterize how the variable transform as we change the scale.

$$
u'_i = \lambda_i u_i \equiv b^{y_i} u_i, \qquad b^d = \text{number of spins per block}. \tag{29}
$$

where d is the spatial dimension. Then we can now see how different parameters behave under the RG transformation.

- If $y_i > 0$, then u_i grows under RG and the corresponding operator is relevant.
- If $y_i < 0$, then u_i shrinks under RG and the corresponding operator is irrelevant.
- If $y_i = 0$, then u_i is unchanged under RG and the corresponding operator is marginal. If the deformation of

This is in fact related to the QFT picture, for a operator that perturb the fixed-point action is $\sum_i c_i \mathcal{O}_i$, the scaling dimension of $\mathcal{O}_i$ is:

$$\Delta_i = d - y_i. \tag{30}$$

1.4 Scaling Hypothesis

Most observable in StatMech system can be given from the free energy. Thus the behavior of the free energy under RG transformation near a fixed point is of great interest, it may give us behavior of observables near the critical points in terms of the couplings. The free energy density is defined as:

$$f = -\frac{1}{\beta N} \log Z, \tag{31}$$

Under RG transformation, the partition function is invariant, thus the free energy density transforms as:

$$f' = -\frac{1}{\beta N'} \log Z = b^d f \tag{32}$$

remember that b^d is the number of spins per block. The free energy density can be split into:

$$f(\{K, c\}) = c + \hat{f}(\{K\}), \tag{33}$$

We split out the vacuum energy density c since it is irrelevant to most observables. In general we have:

$$c' = b^d(c + g(\{K\})) \tag{34}$$

Thus under RG transformation, we have the following relation:

$$\hat{f}(\{K\}) = g(\{K\}) + b^{-d}\hat{f}(\{K'\}) \tag{35}$$

We now split the free energy density into singular and regular parts. We mostly care about the singular part since it is the one that gives us the signal of the second order phase transition. Normally, the function $g(\{K\})$ is regular and thus under RG the singular part of the free energy density transforms as:

$$f_s(K) = b^{-d} f_s(K'). \tag{36}$$

Now we focus on the case that we have a fixed point K^* and we are **near the fixed point**. We assume now we only have thermal and magnetic scaling variables u_t and u_h, (Just assume other irrelevant variables goes to 0 near the fixed point), we have:

$$f_s(u_t, u_h) = b^{-d} f_s(b^{y_t} u_t, b^{y_h} u_h). \tag{37}$$

After n iterations,

$$f_s(u_t, u_h) = b^{-nd} f_s(b^{ny_t} u_t, b^{ny_h} u_h). \tag{38}$$

Choose to end on some n so that $|b^{ny_t} u_t| = u_{t0}$. Then plugging in n in terms of other quantities gives:

$$f_s(u_t, u_h) = \left|\frac{u_t}{u_{t0}}\right|^{\frac{d}{y_t}} f_s\left(\operatorname{sgn}(u_t)u_{t0}, u_h\left|\frac{u_t}{u_{t0}}\right|^{-\frac{y_h}{y_t}}\right). \tag{39}$$

Using reduced temperature $t = \frac{T-T_c}{T_c}$ and the relation $u_t = \frac{t}{t_0}, u_h = \frac{h}{h_0}$ (we here won't justify why this is true, one can assume as a fact for Ising), this is commonly written as

$$f_s(t, h) = \left|\frac{t}{t_0}\right|^{\frac{d}{y_t}} \Phi\left(\left(\frac{h}{h_0}\right)\left|\frac{t}{t_0}\right|^{-\frac{y_h}{y_t}}\right). \tag{40}$$

Consequently, taking derivatives gives the behavior of other observables:

$$C \sim \partial_t^2 f_s \sim |t|^{-\alpha}, \quad M \sim \frac{\partial f_s}{\partial h}\,|_{h=0} \sim (-t)^{\beta}, \quad \chi \sim \frac{\partial M}{\partial h}\,|_{h=0} \sim |t|^{-\gamma}, \tag{41}$$

with

$$\alpha = 2 - \frac{d}{y_t}, \qquad \beta = \frac{d - y_h}{y_t}, \qquad \gamma = \frac{2y_h - d}{y_t}. \tag{42}$$

We can see that the critical exponents are determined by y_t and y_h, which are determined by the scaling properties of the coupling near the fixed point. And they are related to each other, cause they are not independent.

In a fixed spatial dimension, the Ising lattice model, uniaxial magnets, the liquid–gas critical point, and one-component scalar φ^4 theory can belong to the same Ising universality class, that flow to the same fixed point at IR and thus have the same critical behavior.

2 Conformal Transformations

We now focus on a theory on the RG fixed point. One can deduce that the RG fixed point definition of the correlation function will lead to the following identity:

$$\langle \mathcal{O}_1(\lambda x_1)...\mathcal{O}_n(\lambda x_n)\rangle = \lambda^{-\Delta_1-...-\Delta_n}\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle. \tag{43}$$

One may find that this is exactly the definition of symmetry under scaling transformations of certain covariant fields. In fact, at RG fixed points we often have a larger symmetry group, the conformal group. In this section we will discuss the conformal group and its implications for correlation functions.

2.1 Conformal transformations

2.1.1 The conformal Killing equation

A conformal transformation is a coordinate transformation $x \to x'$ for which

$$g'_{\mu\nu}(x') = \Omega(x) g_{\mu\nu}(x). \tag{44}$$

> **Remark**
>
> This is a very generalized equation, without having the metric fixed. Starting now, we will fix a metric and use the flat Euclidean metric $\eta_{\mu\nu} = (1, 1, 1, ...)$, which is the identity matrix. Notice that the Minkowski metric will also lead to a similar result.

For an infinitesimal map

$$x'^{\mu} = x^{\mu} + \varepsilon^{\mu}(x), \tag{45}$$

the Jacobian and its inverse are

$$\frac{\partial x'^{\mu}}{\partial x^{\alpha}} = \delta^{\mu}_{\alpha} + \partial_{\alpha}\varepsilon^{\mu}, \qquad \frac{\partial x^{\mu}}{\partial x'^{\alpha}} = \delta^{\mu}_{\alpha} - \partial_{\alpha}\varepsilon^{\mu}. \tag{46}$$

To first order, we have:

$$g'_{\mu\nu} = g_{\mu\nu} - \partial_{\mu}\varepsilon_{\nu} - \partial_{\nu}\varepsilon_{\mu}. \tag{47}$$

Writing $\Omega = 1 - f$, one obtains the conformal Killing equation

$$\partial_{\mu}\varepsilon_{\nu} + \partial_{\nu}\varepsilon_{\mu} = f(x) g_{\mu\nu}, \qquad f(x) = \frac{2}{d}\partial_{\mu}\varepsilon^{\mu}. \tag{48}$$

In flat space, differentiating and permuting indices gives

$$2\partial_{\mu}\partial_{\nu}\varepsilon_{\rho} = \eta_{\rho\nu}\partial_{\mu}f + \eta_{\rho\mu}\partial_{\nu}f - \eta_{\mu\nu}\partial_{\rho}f. \tag{49}$$

Tracing and differentiating imply

$$2\partial^2\varepsilon_{\rho} = (2-d)\partial_{\rho}f, \tag{50}$$

$$(2-d)\partial_{\mu}\partial_{\nu}f = \eta_{\mu\nu}\partial^2 f. \tag{51}$$

Hence

$$\begin{cases} \text{no local constraint on } f & d = 1 \\ \partial^2 f = 0 & d = 2 \\ \partial_{\mu}\partial_{\nu}f = 0 & d > 2. \end{cases} \tag{52}$$

2.1.2 Finite conformal transformations

For $d > 2$, $f = A + B_\mu x^\mu$. We then take the ansatz of ε^μ given as:

$$\varepsilon^\mu = a_\mu + b_{\mu\nu}x^\nu + c_{\mu\nu\rho}x^\nu x^\rho \quad (2.11) \tag{53}$$

Here $c_{\mu\nu\rho} = c_{\mu\rho\nu}$. For now a_μ, $b_{\mu\nu}$, and $c_{\mu\nu\rho}$ are arbitrary constants. Finally, we can plug this form of ε^μ into the conformal killing equation and require the condition to be satisfied for all x. This leads to the following:

- a_μ is free of constraints, and corresponds to infinitesimal translations.
- $b_{\mu\nu} = \alpha\eta_{\mu\nu} + m_{\mu\nu}$, with $m_{\mu\nu} = -m_{\nu\mu}$. The trace part corresponds to an infinitesimal scale transformation. $m_{\mu\nu}$ corresponds to an infinitesimal Lorentz transformation/rotation.
- $c_{\mu\nu\rho} = \eta_{\mu\rho}b_\nu + \eta_{\mu\nu}b_\rho - \eta_{\nu\rho}b_\mu$, for a constant vector b_μ.

The corresponding finite transformations are

$$\begin{aligned} \text{translation:} \quad & x'^\mu = x^\mu + a^\mu, \\ \text{rotation/Lorentz:} \quad & x'^\mu = M^\mu_\nu x^\nu, \\ \text{dilatation:} \quad & x'^\mu = \lambda x^\mu, \\ \text{SCT:} \quad & x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b\cdot x + b^2x^2}. \end{aligned} \tag{54}$$

An important transformation is called inversion is

$$I: \quad x'^\mu = \frac{x^\mu}{x^2}, \tag{55}$$

and an SCT can be written as inversion–translation of $-b$ –inversion.

2.2 Representation

2.2.1 Representation on Functions

Definition 2.1 **Scalar function**

For a scalar function with zero intrinsic weight, we define it as transforming under the following rule with respect to any spacetime transformation $x \to x'$:

$$f'(x') = f(x), \tag{56}$$

Definition 2.2 **Generator**

For a transformation acting on a function, a generator G_a is defined by following with some real parameter ω_a:

$$f'(x) = e^{-i\omega_a G_a} f(x). \tag{57}$$

For a scalar function the generator for arbitrary transformation can be easily derived: We consider an infinitesimal version of a transformation, which is given by:

$$x'^\mu = x^\mu + \omega_a A^\mu_a(x), \qquad A^\mu_a(x) = \frac{\partial x'^\mu}{\partial \omega_a}\,|_{\omega=0}, \tag{58}$$

then we have:

$$\begin{aligned} f'(x) - f(x) &= -i\omega_a G_a f(x) = f(x - \omega_a A_a(x)) - f(x) \\ &= -\omega_a A_a^\mu(x)\partial_\mu f(x). \end{aligned} \tag{59}$$

$$iG_a f(x) = A_a^\mu(x)\partial_\mu f(x). \tag{60}$$

Then with certain conventions, one can easily find the generators of the conformal group on a scalar function:

$$\begin{aligned} P_\mu &= -i\partial_\mu, \\ L_{\mu\nu} &= i\big(x_\mu\partial_\nu - x_\nu\partial_\mu\big), \\ D &= -ix^\mu\partial_\mu, \\ K_\mu &= -i\big(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\big). \end{aligned} \tag{61}$$

Notice that the definition of the generators are convention dependent, however, $\omega_a G_a$ is convention independent. To be more concrete, we choose the convention that ω_a is related to the parameters of the finite conformal transformation as:

$$\omega_a G_a = a^\mu P_\mu + \frac{1}{2}\theta^{\mu\nu}L_{\mu\nu} + \lambda D + b^\mu K_\mu, \qquad \theta^{\mu\nu} = -\theta^{\nu\mu}. \tag{62}$$

The parameters have the following geometric meanings:

- a^μ is the translation distance: $x'^\mu = x^\mu + a^\mu$.
- $\theta^{\mu\nu}$ specifies an infinitesimal rotation in Euclidean spacetime, or a rotation/boost in Minkowski spacetime. Its related to the coordinate transformation by the canonical exponential map:

$$x'^\mu = e^{\frac{1}{2}\theta^{\rho\sigma}L_{\rho\sigma}}x^\mu \quad L_{\rho\sigma} = x_\rho\delta^\mu_\sigma - x_\sigma\delta^\mu_\rho. \tag{63}$$

- λ is the logarithm of the dilation factor: $x'^\mu = e^\lambda x^\mu$.
- b^μ specifies a special conformal transformation. It has the dimension of inverse length. The corresponding finite transformation is

$$x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b_\nu x^\nu + b^2x^2}. \tag{64}$$

Together, these parameters give the infinitesimal coordinate transformation

$$x'^\rho = x^\rho + a^\rho + \theta^{\rho\nu}x_\nu + \lambda x^\rho + 2(b_\nu x^\nu)x^\rho - b^\rho x^2 + O(\omega^2). \tag{65}$$

For example, a finite dilatation gives

$$f'(r') = e^{-i\alpha D}f(r') = e^{-\alpha r'\partial_{r'}}f(r') = f(e^{-\alpha}r') = f(r), \tag{66}$$

where $r' = e^\alpha r$. by exponentialing the derivative we mean it by Taylor expansions.

2.2.2 Conformal algebra

With the conventions above, we can find an algebra of the conformal generators.

$$\begin{aligned} \big[D, P_\mu\big] &= iP_\mu, \quad \big[D, K_\mu\big] = -iK_\mu, \quad \big[K_\mu, P_\nu\big] = 2i\big(\eta_{\mu\nu}D - L_{\mu\nu}\big), \\ \big[L_{\mu\nu}, P_\rho\big] &= -i\big(\eta_{\mu\rho}P_\nu - \eta_{\nu\rho}P_\mu\big), \quad \big[L_{\mu\nu}, K_\rho\big] = -i\big(\eta_{\mu\rho}K_\nu - \eta_{\nu\rho}K_\mu\big), \\ \big[L_{\mu\nu}, L_{\rho\sigma}\big] &= -i\big(\eta_{\mu\rho}L_{\nu\sigma} - \eta_{\mu\sigma}L_{\nu\rho} - \eta_{\nu\rho}L_{\mu\sigma} + \eta_{\nu\sigma}L_{\mu\rho}\big). \\ \big[P_\mu, P_\nu\big] &= \big[K_\mu, K_\nu\big] = [D, D] = \big[D, L_{\mu\nu}\big] = 0, \end{aligned} \tag{67}$$

Defining

$$J_{\mu\nu} = L_{\mu\nu}, \quad J_{-1,\mu} = \frac{1}{2}(P_\mu - K_\mu), \quad J_{0,\mu} = \frac{1}{2}(P_\mu + K_\mu), \quad J_{-1,0} = D, \tag{68}$$

one recognizes the algebra $\mathrm{so}(d+1,1)$ in Euclidean signature. If we consider Minkwoski spacetime with signature $(-,+,+,+,...)$, the algebra is $\mathrm{so}(d,2)$.

> **Remark**
>
> Above discussion is all in Euclidean spacetime, however, one can prove that the Minkowski version is just changing the metric.

> **Remark**
>
> Why functions instead of the defining representation? This question in fact we have encountered in Poincare representation. Its hard to get the commutation of the spacetime translation and the lorentz transformation using the defining representation. Thus we often just use representation on functions or an abstract representation on Hilbert space to get the algebra.

2.2.3 Action on local operators

Now we consider the action of the conformal group on the Hilbert space. For a quantum field theory, the symmetry transformations are represented by unitary operators on the Hilbert space (and generators are Hermite operators, at least for Lorenzian Theory a theory in Euclidean spacetime may be more subtle to consist with the Wick rotation).

> **Remark**
>
> In fact the Hermite conjugation's definition is dependent on the quantization. For lorentizan theory, we usually don't have this subtleties (one may think of Unrul effect in Rindler spacetime but this is not quite a same problem.). In Euclidean signature, different choice of time will lead to different consist definition of Hermite conjugation and in this case we may not really have the generators to be Hermite.
>
> Here we naively assume unitary, we can think of this as we're working in Lorentzian theory.

Consider a field that transforms under some transformation as:

$$x' = F(x), \qquad \varphi'(x') = F(\varphi(x)), \tag{69}$$

> **Definition 2.3** **Quantum generator**
>
> For a quantum operator, it is assumed to be unitary operator and it act on local fields as:
>
> $$\varphi'(x) = U^{-1}\varphi(x)U, \qquad U = e^{-i\omega_a \hat{G}_a}. \tag{70}$$
>
> Now $\hat{G}_a$ is the quantum generators of the theory and is assumed to be a Hermite operator.

Remark

A more general definition is that the quantum generators transform local operators as:

$$\varphi'(x') = O^{-1}\varphi(x)O, \qquad O = e^{-i\omega_a \hat{G}_a}. \tag{71}$$

where O may not be unitary under certain quantization time choice.

Here we will define ω_a just as the parameters we used for the transformation on scalar functions.

Remark

This relation should be understood as a definition for a quantum generator. In QFT canonical quantization scheme we often find the quantum generators are the conserved charges exactly which lead to this convention.

The infinitesimal action WLOG can contain both an orbital part and a finite-dimensional spin part:

$$\left[\hat{G}_a, \varphi(x)\right] = -G_a\varphi(x) + iR(G_a)\varphi(x). \tag{72}$$

For the momentum generators, we have:

$$\varphi'(x) = e^{ia\cdot\hat{P}}\varphi(x)e^{-ia\cdot\hat{P}}, \quad \varphi'(x) = \varphi(x-a). \quad \varphi(x+a) = e^{-ia\cdot\hat{P}}\varphi(x)e^{ia\cdot\hat{P}}. \tag{73}$$

Thus we have:

$$\varphi(x) = e^{-ix\cdot\hat{P}}\varphi(0)e^{ix\cdot\hat{P}}, \tag{74}$$

Remark

This might look a bit different from the books using $(+,-,-,-)$ conventions, here we are using $(+,+,+,+)$ conventions. This note is self consistent with this convention.

The infinitesimal action of the translation generator is then:

$$\left[\hat{P}_\mu, \varphi(x)\right] = i\partial_\mu\varphi(x). \tag{75}$$

Thus, we may see that we only need to define the action of conformal generators on the local operators at the origin, their commutation relation will tell us how they act on local operators at any points.

2.2.4 Primary and Descendant Fields

Definition 2.4 **Primary Fields**

We define a certain series of local operators φ_α to commute with the generators as:

$$\left[\hat{D}, \varphi_\alpha(0)\right] = i\Delta\varphi_\alpha(0), \quad \left[\hat{L}_{\mu\nu}, \varphi_\alpha(0)\right] = i\left(S_{\mu\nu}\right)_{\alpha\beta}\varphi_\beta(0). \quad \left[K_\mu, \varphi_\alpha(0)\right] = 0. \tag{76}$$

The commutation relations beyond the origin can be derived from the algebra with the translation generator.

$$
\begin{aligned}
\big[P_\mu, \varphi_\alpha(x)\big] &= i\partial_\mu \varphi_\alpha(x) \\
[D, \varphi_\alpha(x)] &= i\big(\Delta + x^\mu \partial_\mu\big)\varphi_\alpha(x) \\
\big[L_{\mu\nu}, \varphi_\alpha(x)\big] &= -i\big(x_\mu \partial_\nu - x_\nu \partial_\mu\big)\varphi_\alpha(x) + i\big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(x) \\
\big[K_\mu, \varphi_\alpha(x)\big] &= 2ix_\mu \Delta\varphi_\alpha(x) + i\big(2x_\mu x^\nu \partial_\nu - x^2\partial_\mu\big)\varphi_\alpha(x) + 2ix^\rho\big(S_{\rho\mu}\big)_{\alpha\beta}\varphi_\beta(x)
\end{aligned}
\tag{77}
$$

One can also derive the finite transformation of the primary field, for example under the dilation transformation, we have:

$$
\varphi'(\lambda x) = e^{i\lambda\hat{D}}\varphi(\lambda x)e^{-i\lambda\hat{D}} = \lambda^{-\Delta}\varphi(x). \tag{78}
$$

For a scalar primary field, the finite transformation under a general conformal transformation is:

$$
\varphi'(x') = \Omega(x)^{-\frac{\Delta}{d}}\varphi(x). \quad \Omega(x) = \det\frac{\partial x'^\mu}{\partial x^\nu} \tag{79}
$$

For a generic scaling field that commute as:

$$
[D, \varphi(0)] = i\Delta\varphi(0). \tag{80}
$$

The algebra shows that P_μ raises the dimension by one, while K_μ lowers it by one:

$$
\begin{aligned}
\big[D, P_\mu\varphi(0)\big] &= i(\Delta+1)P_\mu\varphi(0), \\
\big[D, K_\mu\varphi(0)\big] &= i(\Delta-1)K_\mu\varphi(0).
\end{aligned}
\tag{81}
$$

In a unitary theory the spectrum of dimensions is bounded below. Thus we can see that by acting the P_μ operators on a primary field we can generate a seires of scaling field with highier scaling dimensions:

$$
\big[P_{\mu_1}...P_{\mu_n}, \varphi(0)\big] \sim \partial_{\mu_1}...\partial_{\mu_n}\varphi(0). \tag{82}
$$

We call these fields descendants of the primary field.

2.3 Noether's theorem and the stress tensor

For an action

$$
S = \int d^dx\mathcal{L}(\varphi, \partial_\mu\varphi), \tag{83}
$$

Noether's theorem associates a conserved current to every continuous symmetry. In a general infinitesimal transformation, one convenient form is

$$
j^\mu_a = \left[\frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\partial_\nu\varphi - \delta^\mu_\nu\mathcal{L}\right]\frac{\partial x^\nu}{\partial\omega_a} - \frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\frac{\partial\tilde{F}}{\partial\omega_a}, \quad \partial_\mu j^\mu_a = 0. \tag{84}
$$

The translation current is the canonical stress tensor,

$$
T^{\mu\nu}_{\mathrm{c}} = -\eta^{\mu\nu}\mathcal{L} + \frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\partial^\nu\varphi, \quad \partial_\mu T^{\mu\nu}_{\mathrm{c}} = 0. \tag{85}
$$

By an improvement one may choose a specific symmetric tensor $T^{\mu\nu} = T^{\nu\mu}$. Under a general coordinate transformation $x'^\mu = x^\mu + \varepsilon^\mu(x)$, we have:

$$
\delta S = -\frac{1}{2}\int d^dx T^{\mu\nu}\big(\partial_\mu\varepsilon_\nu + \partial_\nu\varepsilon_\mu\big) \tag{86}
$$

If we choose this general transformation to be a conformal transformation, we have due to the conformal killing equation:

$$\delta S = -\frac{1}{2}\int d^d x f(x) T^{\mu}{}_{\mu} \tag{87}$$

Scale invariance means that $\delta S = 0$ when $f(x) = \text{const}$ this will lead to:

$$T^{\mu}{}_{\mu} = \partial_{\mu} V^{\mu}, \tag{88}$$

with a virial current V^{μ}. If we also have conformal invariance, we have $\delta S = 0$ for SCT, then:

$$T^{\mu}{}_{\mu} = 0. \tag{89}$$

In summary, in CFT we always can have a improved symmetric and traceless stress tensor $T^{\mu\nu}$, which is conserved.

2.4 Implications for correlation functions

For a QFT we define the correlation function as:

$$\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle = \left(\frac{1}{Z}\right)\int D\varphi \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)e^{-S[\varphi]}, \tag{90}$$

notice that we always work in the Euclidean signature. One can prove that for a symmetry we have the result that:

$$\langle \varphi(x_1')...\varphi(x_n')\rangle = \langle \mathcal{F}(\varphi(x_1))...\mathcal{F}(\varphi(x_n))\rangle \tag{91}$$

By $\mathcal{F}(\varphi(x))$ we means the RHS of the definition of field condiguration transformation:

$$\varphi'(x') = \mathcal{F}(\varphi(x)). \tag{92}$$

which is the transformation purely of field (not the spacetime point). Notice that this is pure path integral and have nothing to do with whether the transformation is Unitary in certain time choice or not.

2.4.1 Two-point functions

Translation and rotation invariance imply dependence only on $x_{12} = x_1 - x_2$. Dilatations give

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\rangle = \frac{C_{12}}{(|x_{12}|)^{\Delta_1+\Delta_2}}. \tag{93}$$

Under an SCT,

$$x_{12'}^2 = \frac{x_{12}^2}{\gamma_1\gamma_2}, \qquad \gamma_i = 1 - 2b\cdot x_i + b^2 x_i^2. \tag{94}$$

SCT covariance requires $\Delta_1 = \Delta_2$. In an orthonormal basis of scalar primaries,

$$\langle \mathcal{O}_{i(x_1)}\mathcal{O}_{j(x_2)}\rangle = \frac{C_i\delta_{ij}}{(x_{12}^2)^{\Delta_i}}. \tag{95}$$

2.4.2 Three-point functions

For scalar primaries, conformal symmetry fixes the position dependence completely:

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = \frac{\lambda_{123}}{\left[(x_{12}^2)^{\frac{\Delta_1+\Delta_2-\Delta_3}{2}}(x_{13}^2)^{\frac{\Delta_1+\Delta_3-\Delta_2}{2}}(x_{23}^2)^{\frac{\Delta_2+\Delta_3-\Delta_1}{2}}\right]}. \tag{96}$$

2.4.3 Four-point functions and cross-ratios

For four identical scalar primaries of dimension Δ,

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \frac{1}{\left[(x_{12}^2)^{\Delta}(x_{34}^2)^{\Delta}\right]}, \mathcal{G}(u,v), \tag{97}$$

where

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2} = z\overline{z}, \qquad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2} = (1-z)(1-\overline{z}). \tag{98}$$

Notice that the definition of z and $\overline{z}$ is conventionally not the same for Euclidean and Lorentzian theory (though the corrator will always take the same form).

In Euclidean signature $\overline{z} = z^*$. In Lorentzian signature z and $\overline{z}$ are independent real variables in the standard kinematic region.

A standard Lorentzian configuration is

$$x_1 = 0, \qquad x_2 = (t, x, 0, ...), \qquad x_3 = (0, 1, 0, ...), \qquad x_4 = \infty. \tag{99}$$

Then

$$u = x^2 - t^2, \qquad v = (1-x)^2 - t^2, \qquad z = x - t, \qquad \overline{z} = x + t. \tag{100}$$

Bibliography